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"""
Dieline Generator – a "mini Esko" for
structural packaging design.

Deploy target: Streamlit Community Cloud
(same pattern as PharmaVerify).
Local run:      streamlit run app.py
"""

import streamlit as st
import streamlit.components.v1 as
components

from geometry import CartonSpec,
generate_tuck_end_carton
from export import lines_to_svg

st.set_page_config(page_title="Dieline
Generator", layout="wide")

st.title("📦 Dieline Generator")
st.caption(
    "A parametric structural packaging
    design tool – the core function of "
    "Esko's ArtiosCAD, scoped to one carton
    style: the straight tuck-end (STE) "
    "folding carton used for pharma
    secondary packaging."
)

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# ----- Sidebar: dimensional +
material inputs -----
st.sidebar.header("Carton Dimensions")
L = st.sidebar.slider("Length / front width
(mm)", 20.0, 200.0, 60.0, step=1.0)
D = st.sidebar.slider("Depth / side width
(mm)", 10.0, 150.0, 30.0, step=1.0)
H = st.sidebar.slider("Height (mm)", 20.0,
250.0, 100.0, step=1.0)

st.sidebar.header("Material & Flap
Settings")
thickness = st.sidebar.slider(
    "Board thickness (mm)", 0.1, 1.5, 0.3,
step=0.05,
    help="Drives bulge compensation –
thicker board needs more crease allowance "
    "so the assembled box hits the
target dimensions."
)
glue_flap = st.sidebar.slider("Glue flap
width (mm)", 8.0, 25.0, 15.0, step=1.0)
tuck_depth_ratio = st.sidebar.slider(
    "Tuck flap depth ratio", 0.5, 1.0, 0.8,
step=0.05,
    help="Tuck flap depth as a fraction of
the side panel depth (D)."
)
dust_clearance = st.sidebar.slider(
    "Dust flap clearance (mm)", 0.5, 5.0,
2.0, step=0.5,
    help="How much shorter the dust flaps
sit versus the tuck flaps, so they "
    "don't collide when the carton is

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closed."
)
tuck_taper = st.sidebar.slider("Tuck flap
taper (mm)", 1.0, 10.0, 4.0, step=0.5)

# ----- Generate geometry -----
-----
spec = CartonSpec(
    L=L, D=D, H=H,
    thickness=thickness,
    glue_flap=glue_flap,
    tuck_depth_ratio=tuck_depth_ratio,
    dust_clearance=dust_clearance,
    tuck_taper=tuck_taper,
)
lines = generate_tuck_end_carton(spec)
svg_markup = lines_to_svg(lines)

cut_count = sum(1 for ln in lines if
ln.kind == "cut")
crease_count = sum(1 for ln in lines if
ln.kind == "crease")

# ----- Layout: preview + stats
-----
col_preview, col_stats = st.columns([3, 1])

with col_preview:
    st.subheader("Dieline Preview")
    st.caption("Solid = cut line · Dashed =
crease/fold line")
    # White backing div so the SVG (white
fill) is visible on Streamlit's
    # theme regardless of light/dark mode.

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        components.html(
            f'<div style="background:white;
padding:8px;">{svg_markup}</div>',
            height=500,
            scrolling=True,
        )

with col_stats:
    st.subheader("Dieline Stats")
    st.metric("Cut lines", cut_count)
    st.metric("Crease lines", crease_count)
    flat_width = glue_flap + 2 * L + 2 * D
    + 3 * (2 * thickness)
    st.metric("Flat width (mm)", f"
{flat_width:.1f}")
    st.metric("Flat height (mm)", f"{H + 2
* D * tuck_depth_ratio:.1f}")

    st.download_button(
        label="↓ Download SVG",
        data=svg_markup,
        file_name="tuck_end_carton.svg",
        mime="image/svg+xml",
        use_container_width=True,
    )

# ----- Technical detail
(interview-ready explanation) -----
---
with st.expander("How this dieline is
calculated (technical notes)"):
    st.markdown(
        f"""
Panel sequence: Glue Flap → Front (L)

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→ Side (D) → Back (L) → Side (D),
wrapping into a rectangular tube. The glue
flap glues to the outer face of
the last Side panel to close it.

****Bulge compensation:**** Board doesn't fold
at a mathematical line – it
creases with a radius roughly proportional
to caliper. This design adds
'2 × thickness' to each crease allowance so
the **assembled** box hits your
target L/D/H, not just the flat pattern. At
your current thickness
(*{thickness}* mm), that's a ******
*{2*thickness:.2f}* mm****** allowance per
crease.

****Flap logic:****
- Front/Back get ****tuck flaps**** –
trapezoidal, tapered *{tuck_taper}* mm per
side so they slide into the carton
opening without binding.
- Side panels get ****dust flaps**** – plain
rectangles, sitting
{dust_clearance} mm shorter than the tuck
flaps so they tuck inside
without corner collisions.
- Both top and bottom use the same flap
style (straight tuck), which is
why it's called a Straight Tuck End (STE)
carton – a ****Reverse Tuck End**
(RTE)****** would mirror the bottom flap
direction instead.

****Why this style, not RSC:**** Regular Slotted Containers (RSC) are shipping/transit boxes. Pharma secondary packaging (tablet strips, ampoule cartons) almost always uses tuck-end or reverse-tuck styles – this tool is scoped to match that domain, not general corrugated packaging.

"""

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